

1. a)

An educational management system handles different types of training programs. Write a Java program that performs the following tasks:

- Create an abstract class named `TrainingProgram` with the following:
 - ✓ Two protected data members: `programName (String)` and `sessions (int)`.
 - ✓ A constructor to initialize these variables.
 - ✓ A concrete method `displayProgram()` that prints the program name and number of sessions.
 - ✓ An abstract method `calculateCost()` that returns a double (the cost of training).
- Create a subclass named `Workshop` that extends `TrainingProgram`:
 - ✓ Add an additional data member `sessionFee (double)`.
 - ✓ A constructor to initialize the data members of `Workshop` and its superclass.
 - ✓ Implement the `calculateCost()` method such that the total training cost is:
 $\text{sessions} \times \text{sessionFee}$.
- Create a separate driver class named `TrainingManagement` containing the `main()` method that:
 - ✓ Creates an object of `Workshop` with sample data.
 - ✓ Calls `displayProgram()` and prints the calculated program cost.

b)

A data management system supports different types of storage mechanisms. Write a Java program that performs the following tasks:

- Define an interface named `Storage` that contains:
 - ✓ A method `saveData (String data)` that returns void.
- Create a class named `FileStorage` that:
 - ✓ Implements the `Storage` interface.
 - ✓ Implements the `saveData (String data)` method to display
"Data saved to file: <data>".
- Create a separate driver class named `StorageSystem` containing the `main()` method that:
 - ✓ Creates an object of `FileStorage`.
 - ✓ Uses a `Storage` reference to call the `saveData()` method.

2. A small retail shop records daily item sales quantities in a text file for analysis. Write a Java program that performs the following tasks:

[CO3, C3,
Marks: 20]

- Create a text file named *sales.txt* and write five integer values representing item quantities sold during the day, entered by the user, using the `PrintWriter` class.
- Read the sales data from the file using the `Scanner` class and:
 - ✓ Calculate and display the total number of items sold.
 - ✓ Determine and display the maximum quantity sold in a single entry.
- Use exception handling to:
 - ✓ Handle `FileNotFoundException` if the file cannot be created or opened.
 - ✓ Handle `InputMismatchException` if the user enters invalid (non-integer) input.
 - ✓ Ensure that all file resources are properly closed.

3. A mobile phone contact manager stores contact names using an `ArrayList`. Write a Java program that performs the following tasks:

[CO3, C3,
Marks: 20]

- Create an `ArrayList` named `contacts` to store `String` values representing contact names.
- Add five contact names to the list.
- Display all contact names using a *for-each* loop.
- Search for a contact name entered by the user.
 - ✓ If the contact exists, display a message indicating it was found.
 - ✓ If the contact does not exist, display an appropriate message.
- Display:
 - ✓ The total number of contacts stored.
 - ✓ The first contact in the list (if the list is not empty).